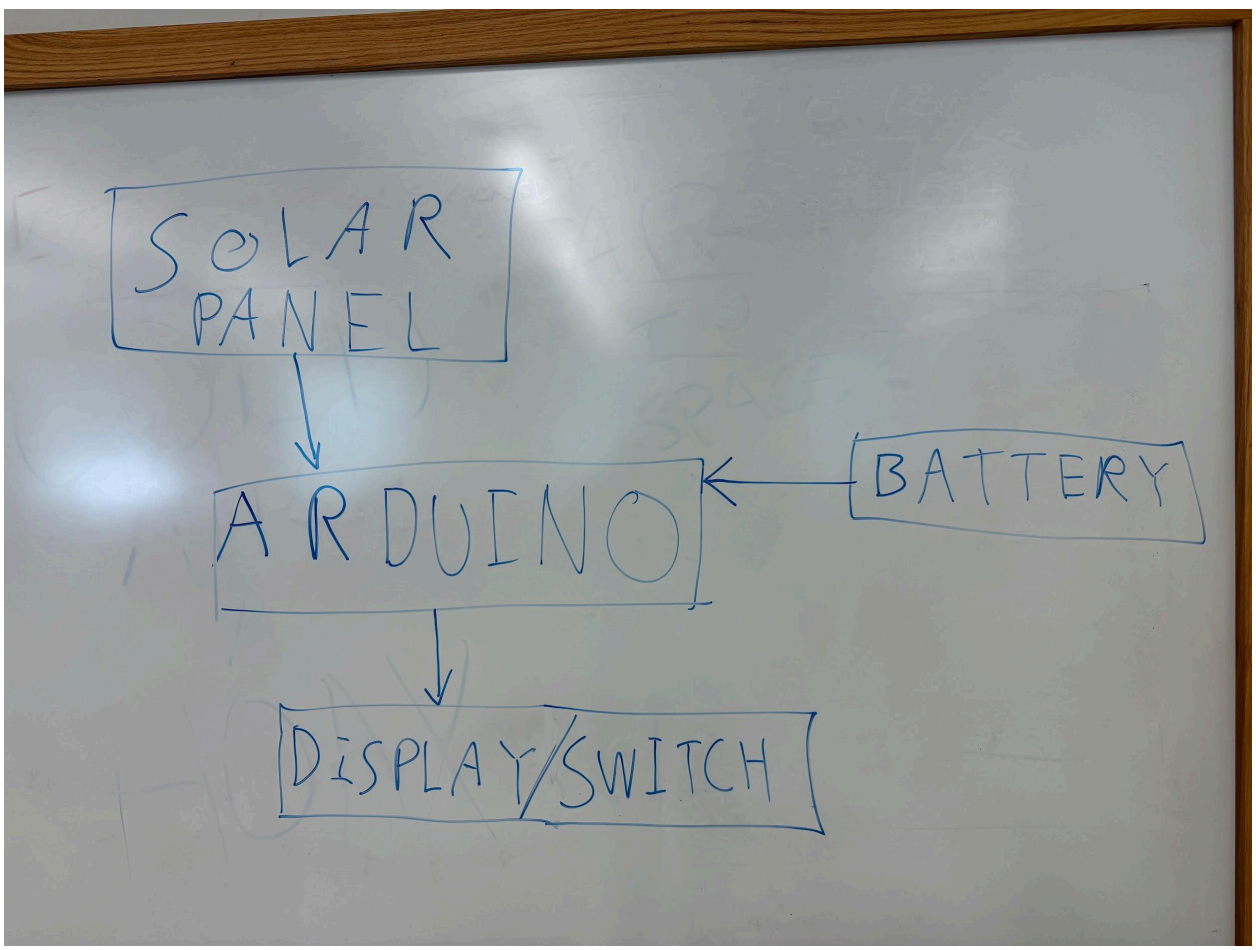This Arduino program implements the control logic for a simple solar charge controller using analog voltage measurements and an OLED display to indicate power routing. The solar panel voltage is monitored on **pin A0**, and the battery voltage is monitored on **pin A1**. The Arduino continuously reads both voltages using its built-in ADC and compares the solar voltage to the required **2.5 V threshold**. If the solar panel voltage is greater than 2.5 V, the program determines that the microcontroller should be powered from the solar panel; otherwise, the battery is selected as the power source. The code also compares the battery voltage to the solar voltage, and if the battery voltage is lower than the solar voltage, it indicates that the battery should be connected to the solar panel for charging. Because the assignment states that no controllable switching element is available for software testing, the actual switching action is not performed in hardware. Instead, the OLED display connected through the Arduino's I2C interface shows the measured solar and battery voltages, the selected microcontroller power source, and whether charging should occur. This allows the Arduino to simulate the decision-making behavior of a solar charge controller while clearly displaying the system's operating state in real time.